

Model Differencing

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Interpretable Model Differencing using JSTs

Introduction

We are trying to compare two ML models that are trained on the same dataset and trying to understand where the models differ in a **human interpretable** way . Traditional metrics like accuracy provide performance measures but they dont give the information about *where* in the feature space models differ. We address this gap through **interpretable model differencing**, which identifies regions of disagreement while contextualizing differences within the models' decision logic.

Problem Formulation

Given two classification ML Models

$$M_1, M_2 : \mathcal{X} \rightarrow \mathcal{Y}$$

and a dataset

$$X \subset \mathbb{R}^{n \times d}$$

, we predict the binary dissimilarity:

$$D(x) = \mathbb{I}[M_1(x) \neq M_2(x)]$$

using an interpretable diff-model

$$\hat{D} : \mathcal{X} \rightarrow \mathbb{R}^+$$

that predicts dissimilarity $D(M_1(x), M_2(x))$ while also being interpretable. The objectives are:

- **Accuracy:** Maximize precision/recall of

$$\hat{D}$$

- **Interpretability:** Represent differences via human-readable rules
- **Contextualization:** Align differences with the models' decision hierarchies

Joint Surrogate Trees (JST)

A **surrogate** is a simpler, interpretable model that approximates a complex model’s behavior. It’s used to explain or analyze the original model’s decisions. Our solution constructs two conjoined decision trees that approximate M_1 and M_2 jointly (Figure 1):

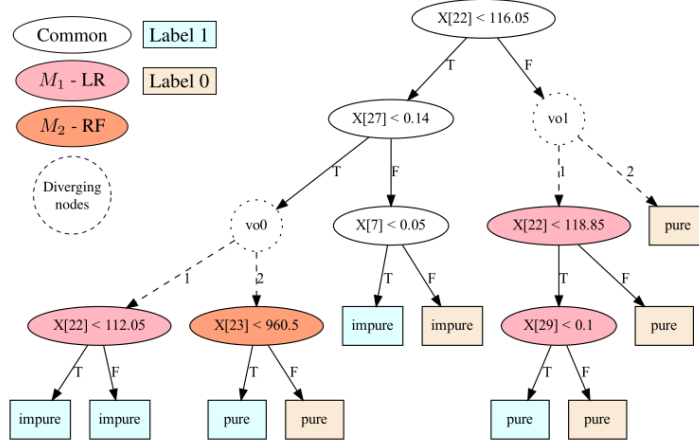


Figure 1: JST structure with shared nodes (white) and model-specific branches (colored). Differences emerge at or-nodes (dotted circles).

Formally, It is defined by the non-empty intersection of path-conditions of differently labelled leaves l_1, l_2 from two decision sub-trees rooted at the same or-node v_o . The collection of all diff regions specifies the diff ruleset:

$$\mathcal{R} = \left\{ pc(l_1) \wedge pc(l_2) : \begin{array}{l} l_i \in leaves(v_o^i), \quad i = 1, 2, \\ label(l_1) \neq label(l_2), \quad v_o \in V_o \end{array} \right\}. \quad (1)$$

where $pc()$ stands for path condition, that is just the *and* of all conditions in the path from the root to the node

Key Properties

1. **Common Prefix:** Shared nodes at upper levels enforce alignment
2. **Divergence Criteria:** Branches split when either model achieves purity
3. **Diff-Rules:** Generated from intersecting leaf pairs with conflicting labels:

IMD: Interpretable Model Differencing

IMD is a technique for comparing two blackbox ML models, M_1 and M_2 , by building a **Joint Surrogate Tree (JST)**. The JST aligns common deci-

sion paths and exposes differences as divergent branches, offering a clear, interpretable comparison.

Traditional methods either compute direct differences or use separate surrogates (which miss alignment). IMD addresses both by using a unified surrogate structure.

IMD proceeds in two main steps:

Step 1: JST Construction

Given input data X , IMD builds a JST using a modified decision tree algorithm. The tree captures:

- **Shared Nodes** that reflect agreement between M_1 and M_2 .
- **Diverging Branches** where the models differ.

From the JST, IMD extracts **diff regions** — input conditions where the models produce conflicting outputs.

Step 2 (Optional): Refinement

If the surrogate predictions in some diff regions don't match the original models, IMD optionally refines the JST locally to improve fidelity.

IMD is model-agnostic, has only one hyperparameter (tree depth), and is simple to implement via small changes to standard greedy decision tree algorithms.

Current Status of Project

- So far, we have read the paper and understood the various concepts presented in it.
- Tried to understand the structure of code given in the official repository.

Further Plan of Action

Further, we would like to do the following :

- Implement individual decision tree surrogates for M_1 and M_2 as a baseline.
- Implement the IMD Algorithm using JSTs as given in the paper.
- Construct the JST for two models as described in the paper.
- Extract and analyze the diff rules from the JST to identify behavioral differences.

- Assess interpretability through qualitative examples and size/depth of the JST.
- Explore hyperparameter sensitivity (e.g., tree depth limits, splitting thresholds).
- Evaluate the fidelity and interpretability of JST compared to individual surrogate trees.
- Explore potential applications or extensions using JST
- Make a detailed report of our findings

Drive link: https://drive.google.com/drive/folders/1cfZu_HhaPS4CR0etYdkVCTINZYfzgJWk